Abdurrahmaan Baghdadi

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Summary

M.S. AI student at UT Austin researching interpretable pathology AI for pediatric brain tumors, while building production LLM and web systems for UT's College of Liberal Arts. 5+ years of research across IoT security, full-stack development, and applied ML; CompTIA Security+ certified.

Education & Certifications

Master of Science (MS): *Artificial Intelligence* | **University of Texas at Austin** Expected May 2027

Bachelor of Science (BS): *Computer Science* | *Minor in Cybersecurity — summa cum laude* | **UTSA** May 2024

3.93 GPA · Dean's List · Honors Program · Distinguished Presidential Scholarship

Certifications: CompTIA Security+ (Jul 2024) | 4 DeepLearning.AI ML specialization courses (2025)

Relevant Coursework: Advances in Deep Learning, Deep Learning, Machine Learning, AI in Healthcare, Algorithms, Operating Systems, Databases

Research Experience

Research Assistant — Pediatric Brain Tumor AI Lab | *University of Texas at Austin (Prof. Ying Ding)* 03/2026 – Present

- Benchmarking pathology foundation models and vision-language models (**UNI**, **CONCH**, **MedSigLIP**, **MedGemma**, **WSI-LLaVA**, **Quilt-LLaVA**) on pediatric brain tumor whole-slide images: zero-/few-shot classification, latent-space and cosine-similarity analysis
- Reproducing PICASSO's TopK **sparse autoencoder** and extending it to compare **UNI2** vs **Virchow2** on ~1,700 TCGA GBM/LGG slides, with patient-level splits, ablations, and **SLURM** pipelines on **TACC** H200 GPU nodes

Professional Experience

Student Technology Assistant | *LAITS, UT Austin College of Liberal Arts* 09/2026 – Present

- Built a helpdesk chatbot router that sends questions to the right one of 64 help pages or to a ticket, using embedding-ensemble retrieval plus a self-hosted LLM reader (**Qwen**, **Ollama**/**vLLM**); cut wrong-page answers from **34%** to **13%** in judged simulations; shipped as a tested **FastAPI** endpoint on **Kubernetes**
- Modernizing college WordPress sites into secure containerized Bedrock builds (**Docker**, **Composer**, **Woodpecker CI**, **Rancher**) and rebuilding faculty research-lab sites on the Labs multisite

Backend Developer | *JIAGIA* 09/2026 – Present

- Developing a **Fastify** + **PostgreSQL** REST API (**TypeScript**, **Zod**) for a research and art content platform, with a private **Railway** database

Code Sensei & Camp Director | *Code Ninjas – Round Rock, TX* 01/2025 – 08/2025

- Grew program sales by **13%** by overhauling operations workflows and targeted outreach; taught up to **15 students** ages 5–14 through structured lessons (**Machine Learning**, **Robotics**, **JavaScript**, **C#**)

Research Assistant — IoT Security | *UTSA – IoT Lab, SCADA Lab* 06/2019 – 01/2025

- Reduced equipment loss by **25%** by building a full-stack inventory system tracking 50–100 IoT devices (**Python/Flask**, **React/TypeScript**, **SQL**)
- Recovered forensic artifacts from IoT devices and automated firmware **NIST** compliance checks (**Autopsy**, **Emba**); captured IoT threat telemetry with honeypots (**ELK**, **ROCK NSM**); secured **9 Linux** VMs and SCADA lab infrastructure

Selected Projects

- **Energy AI Hackathon — Mission Impossible (Jan 2026):** Sole programmer on an interdisciplinary team; ensemble stacking (**HGB/GBR/ExtraTrees/Ridge**) with **GroupKFold CV** and physics-based features; achieved **R^2 = 0.9849** OOF among 40 teams (Phillips66 / Chevron / ExxonMobil sponsored)
- **Sleep Staging from Apple Watch Data: LightGBM** + forward–backward **HMM** smoothing on 247 nights of heart-rate and motion data; raised Cohen's κ in every CV fold (**0.42** → **0.45**), with subject-level bootstrap CIs
- **Advances in Deep Learning — UT Austin:** Implemented half precision, **LoRA**, 4-bit quantization, and **QLoRA** from scratch; built a **BSQ** tokenizer and autoregressive transformer for image generation and compression (**PyTorch**)

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Swift, Java, PHP, C, C#, Bash, R, HTML/CSS
ML & AI: PyTorch, Hugging Face, LoRA/QLoRA, RAG, vLLM, Ollama, LightGBM, XGBoost, Scikit-learn, Pandas
Web, Mobile & Data: FastAPI, Fastify, Flask, React, Node.js, WordPress, SwiftUI, PostgreSQL, MySQL, MongoDB
Infra & Security: Docker, Kubernetes/Rancher, SLURM/TACC HPC, GitHub Actions, AWS EC2, Linux, Wireshark, ELK, Autopsy